From DevOps to GovOps:

The Application of Engineering Principles for the Continuous Optimization of Regulatory Environments

Executive Overview:

Policymakers are often caught between two competing objectives. On one hand, they have an obligation to protect their constituents. The historical track record on this is clear: we breathe clean air and drink clean water because of regulations, not because industry chose to limit its profits to provide us with those things on its own. On the other hand, policymakers are expected to maintain steady economic growth and employment opportunities, even when these objectives appear to conflict with the protections they are responsible for maintaining.

This problem is compounded by the way regulations accumulate. Statutes, administrative requirements, judicial decisions, and agency procedures are often layered onto existing systems over time, without a clear picture of how they interact. For example, a homebuilder might be intimately aware of how many permits, regulations, and administrative hurdles are involved in building a house, along with the time and costs they impose. A lawmaker tasked with creating and maintaining this system often is not. This is not an issue of competence, but rather a lack of the tools needed to provide lawmakers with that visibility.

Thankfully, we now have technology that can allow lawmakers to see the regulatory landscapes they produce. This proposal illustrates how that is possible: giving lawmakers the ability to compare and optimize protective outcomes, end-to-end processing times, associated costs, and even litigation risk, while automatically receiving information about similar regulatory approaches being implemented across the country. The idea is to identify the most efficient combination of protections while allowing people and businesses to move forward quickly, and to give lawmakers the tools they need to accomplish both.

In short: An environmental policy that prevents firms from polluting is successful. A policy that prevents pollution by making it broadly impossible to build anything, anywhere, is not. And lawmakers need tools that make both outcomes immediately legible.

Introduction:

We have different words for it – Vetocracy[1] – Kludgocracy[2] – Everything Bagel governance.[3] But the failure is familiar: Governments increasingly fail at executing on their own goals even when expertise, funding, and clear democratic mandates are all present.[4] The problem is a cumulative one. The state

[1] Fukuyama, Francis. Political Order and Political Decay: From the Industrial Revolution to the Globalization of Democracy. New York: Farrar, Straus and Giroux, 2014. pg. 514–517.
[2] Teles, Steven M. Kludgeocracy: The American Way of Policy. Washington, D.C.: New America Foundation, 2012. pg. 5–6.
[3] Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 113.
[4] Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 71–78.

has accumulated layers of statutes, regulations, court decisions, and agency procedures as a defense for past abuse, but the result is a regulatory system that no institution can fully read, parse or maintain.

While each of these reforms made sense in isolation, their aggregate effect has brought operational throughput to a crawl[5], and costs have exploded relative to peer countries.[6] The political debate has often defaulted to a dichotomy between maintaining the opaque and systemic 'kludge' as it was implemented, or broadly removing safeguards writ large without a clear system for understanding which ones serve a legitimate public purpose, and which ones just increase the bureaucratic lift required to accomplish a task.

This proposal supplies the missing maintenance layer for these systems. It links the legal text of a regulation with a direct operational schema to observe how that text is translated to agency procedures. It enables direct comparison of both protective indices (e.g. clean air, clean water), operational throughput, and legal text across state and local jurisdictions. It proposes a regulatory sandbox structure that compares a proposed change with matched controls. And finally, it maintains it all in a machine readable and version controlled environment that enables line-by-line traceback to which lawmakers proposed which proposal, with merge packets detailing the exact changes in throughput and protective indices.

None of this involves fundamentally new or unproven technology. The proposal combines capabilities already used across the RegTech industry, regulatory information management, software engineering, and data science. Platforms such as Regology[7] and CUBE[8] already structure and compare regulatory text, track compliance and risk, provide Retrieval Augmented Generation (RAG) style AI search services, and help companies map regulatory text to operational workflows. Regulatory information management and PUC e-filing systems provide systems of record for documents, cases, and administrative workflows, while Git-style version control and RAG provide mature models for managing changes and querying complex information. What this proposal adds is a standardized architecture connecting these capabilities at the clause level of each regulation – linking legal text to the workflows and dependencies it creates, providing a common layer and enabling the software ecosystem that has been built on top of these workflows.

1. The Implementation Crisis

At their core, regulations are a set of instructions – building a house requires permits from Agency X – building a factory requires approval from Agency Y – wastewater cannot be disposed without meeting standard Z. This is the critical software that governs how things get done in the real world.

But unlike modern data and software systems which enable the full spectrum of 'Create', 'Read', 'Update', and 'Delete' functionality, most state legislatures are effectively only operating at a 'Create' and 'Delete' level of functionality, as they lack the systems to fully visualize the systems they produce. Similar issues have been solved in the private sector, as Big Tech companies have utilized mono-repositories to govern systems that serve billions of active users and are continuously worked on by

[5] 5. Teles, S. M. (2013). "Kludgeocracy in America." National Affairs, 17, 97–114.
[6] 6. Klein, Ezra, and Derek Thompson. Abundance. Avid Reader Press, 2025. pg. 77.
[7] Regology, "Regulatory Intelligence Platform", https://www.regology.com/.
[8] CUBE, "RegPlatform", https://www.cube.global/products/regplatform.

thousands of engineers simultaneously. They accomplish this complexity management through systems that enable structured representations, dependency mapping, observability, version control, controlled experimentation, and continuous improvement.

This proposal adapts those disciplines to government administration with the objective being: make systems observable, testable, auditable, and revisable. As a design constraint disclaimer – The legal text remains the law. The dual schema models are traceable workflow projections, but changing the underlying legal text still requires approval from lawmakers. Furthermore, a sandbox produces evidence rather than law, and deployment still requires approval from an applicable authority.

Importantly, this proposal does not require an invasive IT infrastructure overhaul to get started, as this system can start with a pilot of one state-level domain, using pre-existing PDF text extraction and existing document stores. Doing so will open up sandbox functionality to lawmakers immediately, while building additional functionality out as the architecture proves itself.

2. Dual-Schema Regulatory Architecture

This proposal represents every regulatory object through two linked representations. The legal schema contains the legal text of the legal object (be that regulation, order, or judicial interpretation) and is prepended with structured metadata that codebase management software can use to categorize the document by jurisdiction, relevant agency, effective date, scope, amendments, dependencies, etc.

The workflow schema connects to the first schema via foreign key and models the implementation details it produces (be that stages, documentation, agency reviews, inspections, permits, appeals, etc.). Persistent identifiers link each clause of legal text to its consequential workflow implementation.

This connection is maintained by representing the legal text as a machine-readable (markdown) document. Each legal document is stored in a repository that allows broader analysis, and each document contains a YAML header that allows software to recognize, organize, and understand it.

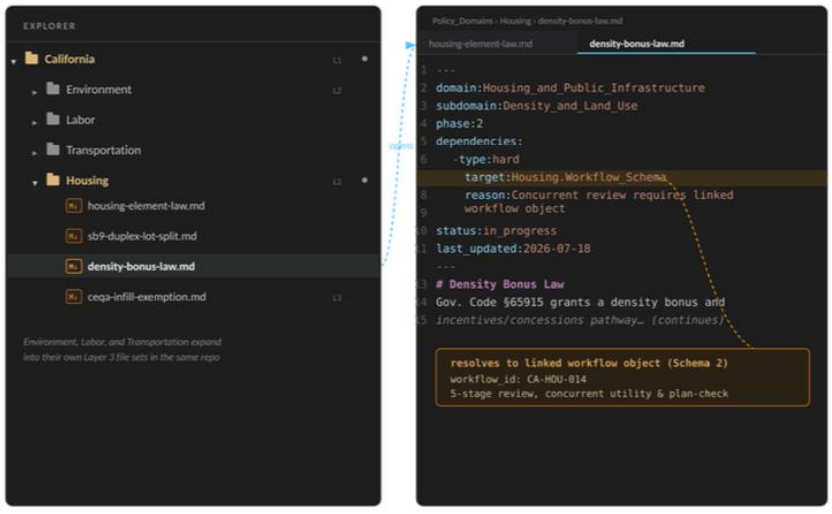


Figure 1 – Repository Organization: *Each policy document is stored in a folder directory regarding its jurisdiction and domain. Users would be able to navigate the directory to quickly find the legal text of Colorado's environmental policy, for example.*

And this YAML frontmatter contains the necessary document classification information for each legal document, along with information regarding dependencies, status, legal identification, and so on. An example could look like the following:

```yaml
---
domain: Housing-and-Urban-Development
subdomain: Density-and-Urban-Zoning
phase: 2-Structurally-Complete

jurisdiction: Colorado
legal-authority: Statute
effective-date: 2026-08-01

document-id: CO-HUD-ZONING-001
schema-version: "1.0"

dependencies:
  - type: Soft
    target: Department-of-Data-and-Accountability.Urban-Density-Zoning-Overlays
    reason: Legal infrastructure for zoning overlay needs to be validated before implementation.

---
```

***Figure 2 – YAML** metadata example.*

This YAML structured text on the head of each document is modeled after open source documentation standards already in use in tech companies and enables database-style operations to quickly compare the legal text between jurisdictions like California, Alabama, and Colorado. Similar strategies combined with existing statutory and regulatory clause identifiers can also enable the clauses of each document to be explicitly linked to the operational workflows they create. Automated version-control hooks can track changes to these provisions, update references when clauses are renumbered, and flag substantive changes or broken dependencies for review.

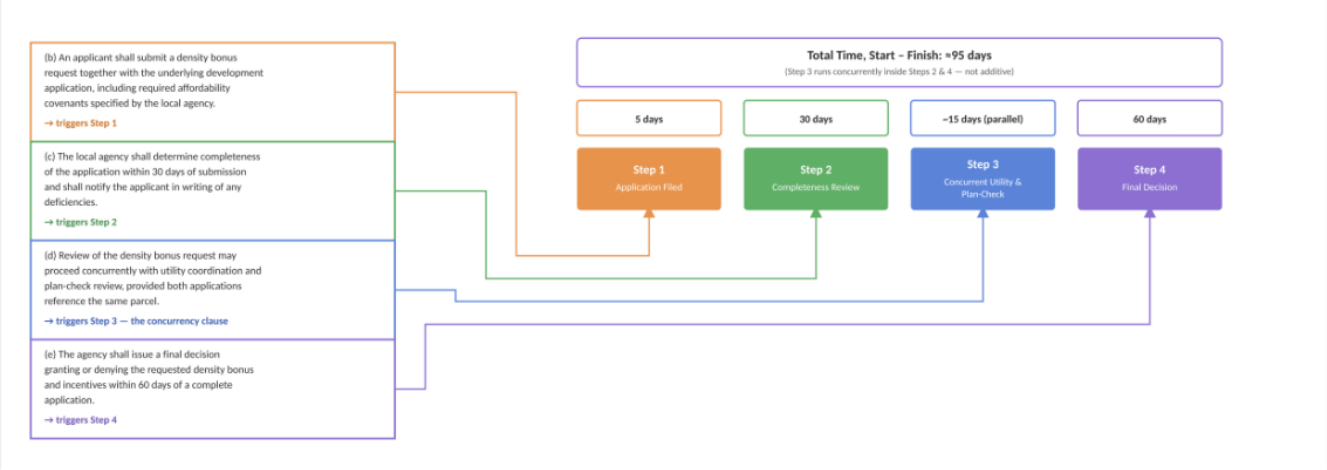


***Figure 3 - Dual Schema Representation:** Figure represents how YAML front matter can be used to explicitly link each clause of legal text to the workflow stage that it creates.*

When implemented on a national scale, this standard allows local governments to maintain their own repositories, contract with their own vendors (or build their own systems on top) while maintaining a common architecture and APIs enabling each government to compare regulatory landscapes, operational throughput, and version histories across jurisdictions. This step also enables higher-level comparative functions.

3. Workflow Intelligence and Regulatory Comparison

The first of the comparative functions that this proposal enables is the direct comparison of operational systems that jurisdictional legal text produces.



Figure 4 – Regulatory Workflow Comparison: *This figure represents the temporal regulatory workflow comparison of state-level jurisdictions for an example infrastructure project.*

Imagine the regulatory approvals needed to build a solar farm – the permits (and their applicable timelines) needed in order to accomplish the work. In this example, California represents the baseline. Colorado preserves an identical sequence of regulatory steps while completing each stage more efficiently, thus completing the process with 60 days to spare. Connecticut, despite taking longer than Colorado on each individual step, executes steps 1 and 2 concurrently and thus shaves 20 days off of the regulatory approval time needed in Colorado (despite each step taking longer in isolation). Texas follows a fundamentally different strategy by omitting intermediate review stages entirely. This turns their regulatory regime into a two-step process that comprises what would be step 1 and step 4 in any other regulatory regime measured. This represents the 'deregulation' approach, which shaves the regulatory approval process down to 45 days – the shortest on this list – but also comes at the cost of a lower protective index.

This illustrates the type of analysis that this proposal can produce. It distinguishes between faster execution of the same protections, concurrent sequencing of independent reviews, removal of duplicated steps, and the removal of legitimate protection. In this instance, California and Colorado can learn from Connecticut's concurrent sequencing of steps 1 and 2. Likewise, Connecticut and California can learn from Colorado's executional efficiency, which allows it to execute the same steps in less time. Each state can learn from Texas if the removal of intermediary steps does not translate to a substantive loss in protections.

This turns the classic "more versus less regulation" debate into a diagnosis of stages, dependencies, handoffs, durations, and outcomes, all the while giving lawmakers traceability into the applicable letter

of the law, and when combined with modern data science concepts, it allows lawmakers to optimize for multiple ‘best outcomes’ at once.

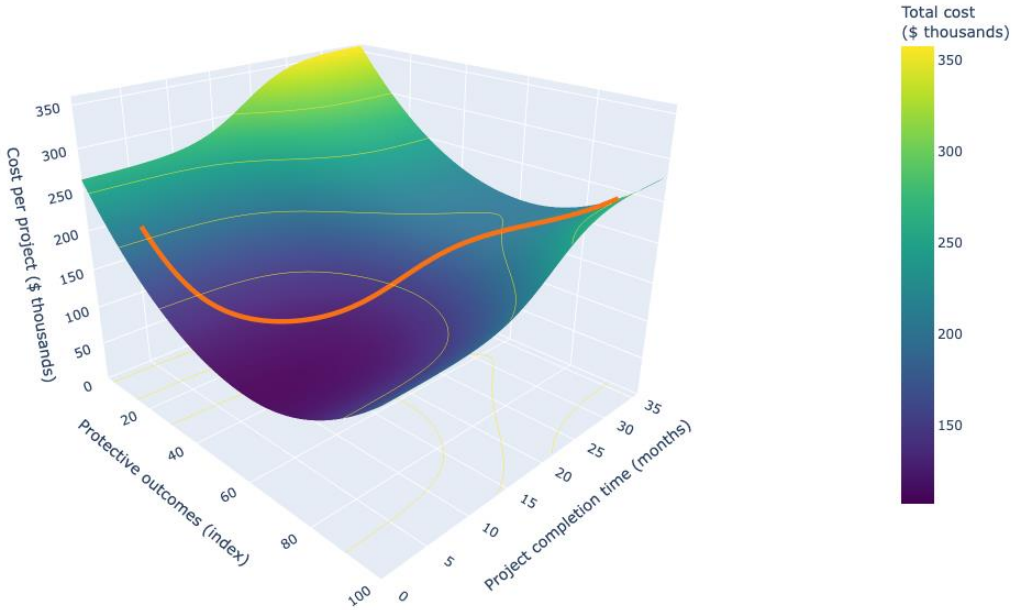


Figure 5 – Calculating the ‘Regulatory Cost’ Landscape: *Here is a simple illustration of how costs and throughput can change as new regulations are introduced. On one axis, costs initially fall as basic protections reduce litigation risk, then start increasing again as accumulated regulatory protections beyond a certain point interact with each other to add veto points, persistent project delays, and even litigation risks and costs (as modeled in the Vetocracy thesis).*[9]

As Figure 5 illustrates, the optimal regulatory framework is not necessarily the blanket deregulation that industry often advocates for. While a maximally deregulated environment can increase speed, it can also increase litigation risk as harms are propagated throughout the community. Conversely, additional regulatory requirements can increase project completion times and associated costs as procedural steps and veto points accumulate. While a generic ‘protective outcomes’ index is used here as a stand-in, such a measure would be calculated from domain-specific, empirical indicators (OSHA-style incident rates for regulations affecting job-site safety, water quality sampling data for clean water, etc.). Each regulatory category would have its own index, derived from one or multiple protective outcomes, to measure against permitting costs and regulatory throughput.

Another important insight from Figure 5 is that **more regulation is not always better either**. Adding additional veto points beyond a critical mass can increase costs and eventually produce a California paradox, where environmental regulations become a highly utilized weapon that special interests can use to veto projects or hold them in litigation indefinitely. Under this scenario, environmental regulations can ironically hurt the environment itself, as green projects are vetoed or delayed by the very regulations they are supposed to supplement. The best strategy is therefore an optimization problem whereby states pursue protections while accurately measuring their benefits in outcomes, while also keeping track of the cost and step-time landscape they produce in aggregate.

[9] Fukuyama, Francis. Political Order and Political Decay: From the Industrial Revolution to the Globalization of Democracy. New York: Farrar, Straus and Giroux, 2014. pg. 540.

Importantly, this method might also surface two different jurisdictions with nearly identical regulatory text, but different timeline and cost outcomes. This would be a flag to the less efficient jurisdiction to overhaul its department and efficiency operations, while being able to keep its textual regulatory protections. Crucially, both inefficient regulatory text, and inefficient implementation outcomes (which were previously obscured) are surfaced by this approach, at little cost to the policymaker.

While this gives lawmakers greater legibility into how they can improve the efficiency of their regulatory regimes, it does not, by itself, give states a way to test new regulatory methods in ways that are safe, bounded, and produce the data they need to make informed decisions. This is detailed in Part 4.

4. Experimental Governance: The Scientific Method, Applied to Government Systems

Despite the legibility offered by the systems above, experiments in the real world have established that 'theory' is not entirely consistent with real life. Therefore, lawmakers need a structured and easy-to-use methodology for testing changes to their state's regulatory regime in a way that is (as aforementioned) safe, bounded, and automatically produces the data they need to make informed decisions about scale or sunset.

To respond to this question, the Regulatory Modernization Sandbox (RMS) is proposed. The RMS is a legally authorized 'branch' of the regulatory system, in the same sense that codebase changes are 'branched' and tested in a bounded and isolated environment. Each sandbox can either be administered digitally within the software system itself or (optionally) directly administered by a Regulatory Modernization Corps (RMC) deployment – government personnel that embed alongside the agencies responsible for executing the workflow under study, or to the site of the regulatory change (alongside local stakeholders) to directly observe how changes affect operations on the ground.

This setup ensures that friction points that occur during the process are identified directly and dealt with up front – instead of waiting years for feedback to filter through complaints, hearings, or elections themselves.

Site Selection and Experimental Design

Sandbox jurisdictions are selected for causal identification rather than convenience. As an illustrative walkthrough, consider a Colorado housing-permitting design. Suppose a change to housing regulations is proposed, and lawmakers want to discern how this proposal changes the amount of time it takes to get the permits needed to build a housing development.

Candidate sandbox jurisdictions are selected by comparing features applicable to median permitting durations and permitting volumes in Colorado jurisdictions. From there, sites are selected according to how well they match up to potential control jurisdictions, as identified through Standardized Euclidean Distance, propensity scores, Mahalanobis distance, or synthetic control weights – the specific matching will be chosen according to the best fit for that regulatory domain. As an example:

Jurisdiction	Population	Median Household Income	Population Density	Owner-Occupied Housing	Median Age	Permitting Volume	Median Permitting Duration
Pueblo	112,000	$54,000	3,200	61%	39	*Illustrative*	*Illustrative*
Greeley	111,000	$60,000	3,700	63%	36	*Illustrative*	*Illustrative*
Longmont	101,000	$82,000	3,900	58%	37	*Illustrative*	*Illustrative*
Loveland	82,000	$76,000	3,400	64%	41	*Illustrative*	*Illustrative*

Figure 6a *- The values above are illustrative and are meant to exemplify how candidate sites can be selected using nearest-neighbors matching from candidate metrics. Each row represents a potential treatment or control jurisdiction. Each column represents a standardized demographic, economic, housing-adjacent, or administrative characteristic used in the matching procedure. An applicable model would include a more robust feature space (featuring target-variable-relevant cleaning, feature selection, etc.).*

From there, applicable treatment-control jurisdictional pairs are selected according to their eligibility, using matching algorithms on the feature matrix to find pairs of jurisdictions that are 'close enough' to compare against each other. Statistical power is then evaluated according to the available sample sizes, outcome variability, and expected treatment effects.

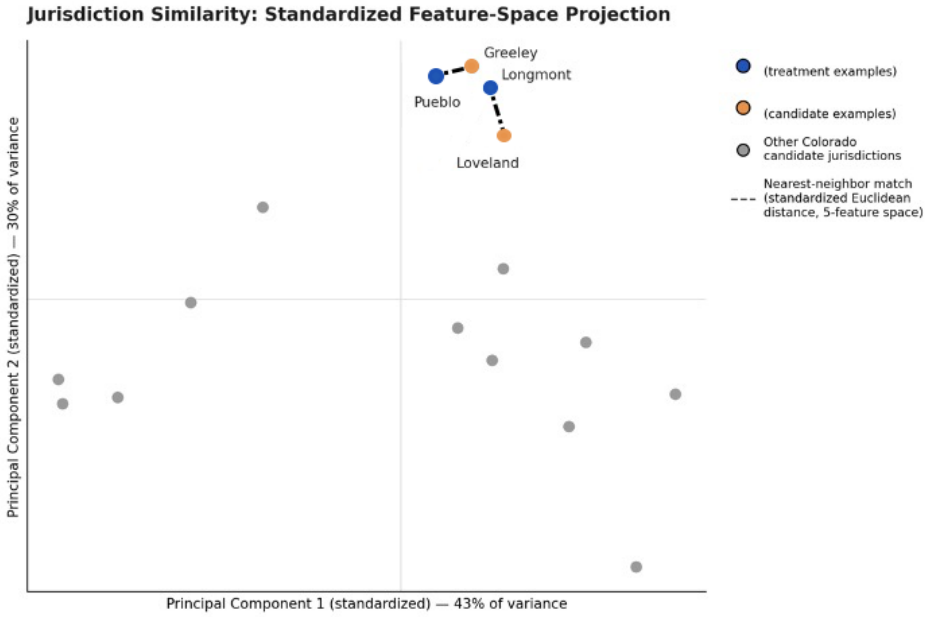


Figure 6b *– Following from the data in Figure 6a, this represents how applicable 'distance' metrics can be used to designate treatment-control pairs. In this case, Greeley and Pueblo are designated as one pair, and Longmont-Loveland as another based on their matched distance calculated from the data in 6a. As a disclaimer, an applicable model would include a more robust feature space (featuring target-variable-relevant cleaning, feature selection, etc.).*

Notably, these matching criteria do not include geographic closeness – the final feature space could be north of dozens of applicable dimensions. Once applicable jurisdictions are selected, they undergo a time-bounded sandbox treatment of potential regulatory changes.



Figure 6c – ***Geographical Distribution:*** *Cities designated as 'treatment' or 'control' are not necessarily geographically close to one another. Instead, pairings are chosen via characteristics such as permitting throughput, population, construction characteristics, etc.*

Evaluation

Once the treatment-control pairs are selected, a Regulatory Modernization Packet (RMP) is prepared to test changes to the regulatory environment and how these changes affect both the median permitting duration as well as the relative protective index. This setup allows lawmakers to test regulatory changes in a bounded manner while minimizing the risk of immediate statewide rollout.

For example, consider a Difference-in-Differences design comparing the median housing permitting time for one matched jurisdictional pair. The line in the middle (designated as T_0) represents the time point in which a regulatory change is made. In this illustration, the 'treatment' jurisdiction (Longmont) experiences a reduction in median permitting time duration relative to its matched control (Loveland).

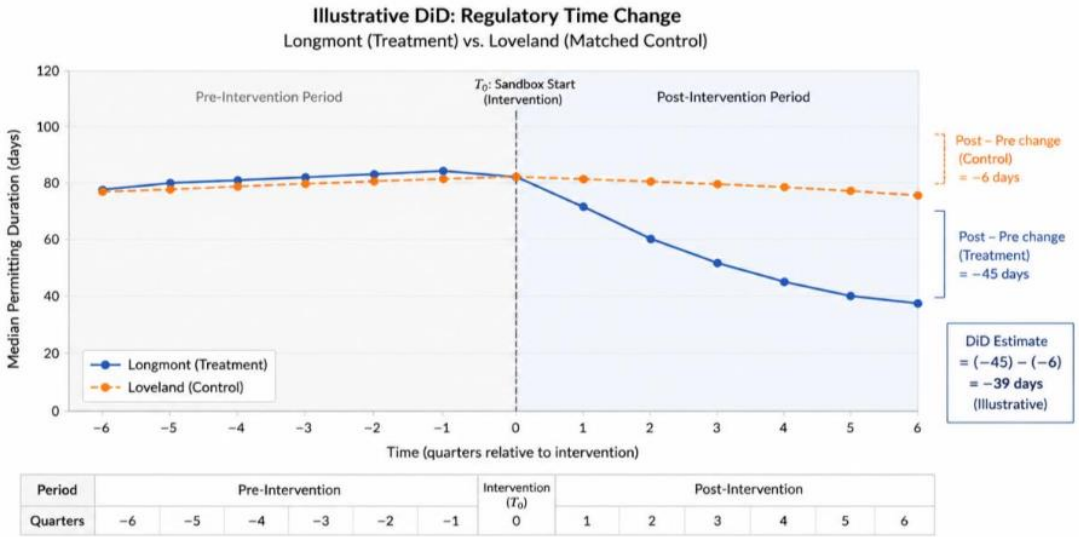


Figure 7 – **Regulatory Change Illustration:** Once a regulatory change is tested, causal analysis will take place comparing the 'treatment' jurisdiction (in this example, Longmont) with the matched control jurisdiction (in this case Loveland).

To increase the statistical rigor, outcomes can also be evaluated using regression discontinuity at jurisdictional boundaries. This is an important step, as geographically separated areas might use different builders, staff, contractors, material suppliers, and so on. The addition of a regression discontinuity design, where boundary conditions permit, allows us to more effectively isolate the effect of the regulatory change. In this example, we'll imagine that Longmont adopts the new regulatory regime, while Firestone and other proximal jurisdictions retain the same regulatory requirements that Longmont previously had. Housing development projects on either side of these jurisdictional boundaries will likely use similar contractors, staff, builders, etc., allowing us to more effectively isolate the effects of the regulatory change.

While multiple experimental designs running concurrently may seem like an over-engineered way to evaluate a regulatory change, it is important to remember that the real world is not a laboratory setting where everything is controlled. Because of this, multiple designs are useful in determining whether the change in permitting time is actually due to the regulatory proposal and not due to the clearance of a local permitting queue, the efficiency of a local contractor, or so on. Incorporating multiple data streams to run these tests is a mature, industry-standard workflow in software environments – and much of the data collection and analysis can be automated. If fully implemented, this proposal would not require expensive personnel deployments to collect the necessary data. That data is an emergent property of the dual schema architecture proposed in part 2, which could be pulled automatically to allow lawmakers to see how that affects permitting throughput from multiple angles.

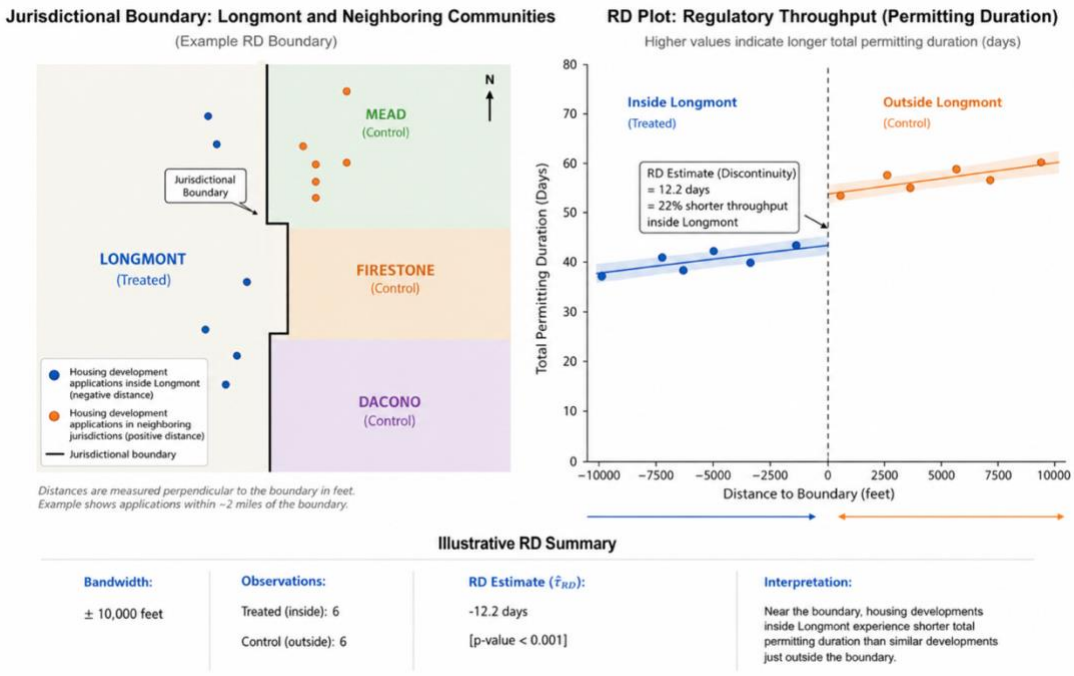


Figure 8 *–* ***Regression Discontinuity Illustration:*** *Once a city is designated as a 'treatment' jurisdiction for a regulatory change, data is automatically pulled for building projects around its periphery. Where boundary conditions permit, this allows researchers to compare permitting outcomes between jurisdictions operating under different regulatory regimes while accounting for differences in local builders, contractors, and suppliers.*

Independent Review and the Dual Key

Importantly, the government cannot be allowed to design, measure, and certify its own experiments – otherwise we are recreating the same 'grade your own work' failure that this setup is meant to avoid. As such, the methods above are combined with a dual-key evidence system built to replicate the peer review structure in scientific fields.

As an overview, this would work by an official evaluating body certifying: baseline conditions, data collection methods, and statistical methods. A separate Independent Research Review panel – composed of accredited universities, nonpartisan research institutions, or other public policy labs – evaluates the site selection, experimental design, data collection, statistical validity, reproducibility, and interpretation of the findings.

Disclaimer: *While these methods are good statistical approximations of a controlled experiment, they are not equivalent to laboratory control. Statistical techniques can construct a counterfactual, and strengthen causal evidence, but they cannot eliminate all potential confounding variables or produce irrefutable conclusions. Those limitations do not constitute an argument against evaluation in this manner. Even imperfect causal measurement is a substantial improvement over the current status quo, in which policies are often implemented without a rigorous mechanism for determining whether they can achieve their intended outcomes.*

5. From Sandbox to Statute

Upon the completion of a successful pilot, the data and pilot results are automatically incorporated into the existing Regulatory Modernization Packet (RMP), which now contains the amendment to the regulatory environment, a 'legal diff' of the exact language change to statute, a 'workflow diff' of the dual-key certified workflow change, and any applicable notes or evidence.

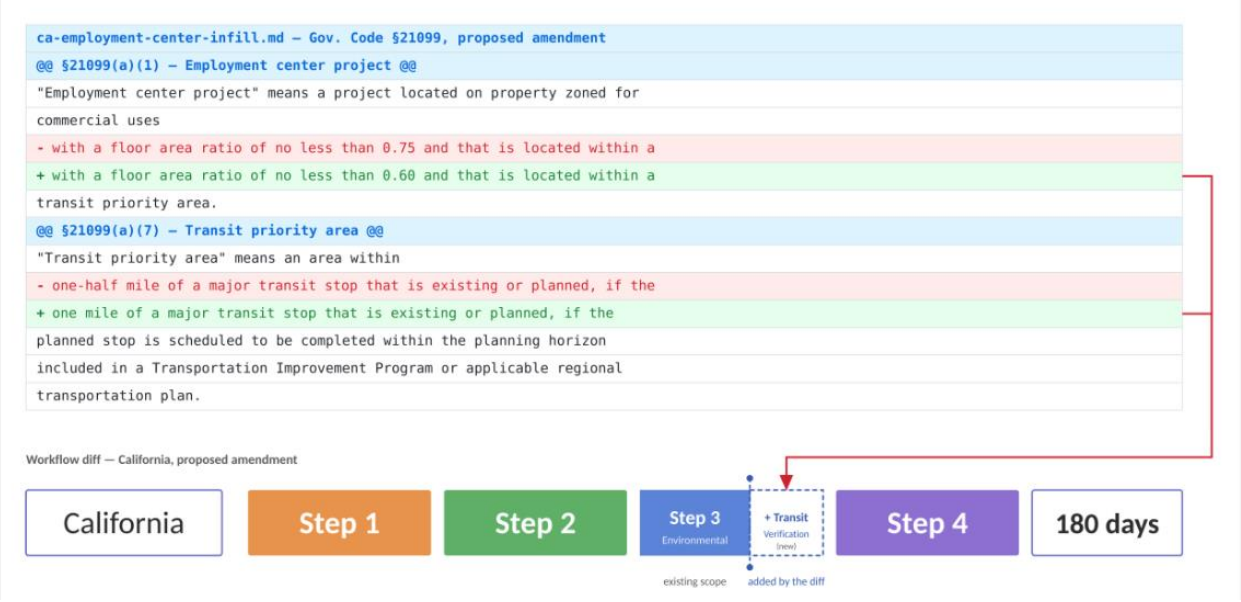


Figure 9 – Regulatory Modernization Packet "Dual Diff" Illustration: *Once up and running, lawmakers can use 'regulatory modernization packets' proposed here to see both the difference in legal text side-by-side with the difference in workflow speed.*

While a successful pilot does not automatically become law, pilot programs can be pre-written by lawmakers to default to either scaling or sunsetting according to independently certified performance metrics. This avoids the traditional trap of successful pilots failing to scale once political headwinds change, or unsuccessful pilots continuing because lawmakers could be embarrassed by the potential news story. Under this setup, if performance metrics are achieved, lawmakers must explicitly intercede to stop a pilot from scaling – or, in the case of the counterfactual, lawmakers must intercede to save a program if it does not meet performance metrics.

In any case, if the applicable legislative body (in this case, the Colorado legislature) approves or has preapproved the changes, the packet containing the changes to the legal and operational schemas merges into the 'production' environment for the broader state. In short: the legislature authorizes the experiment, agencies implement, and independent certification bodies certify performance, which the legislature then either approves or has already preapproved according to the established performance metrics.

6. Incorporating Automation for Long-Term Efficiency

Once the system is set up, the workflow portion of the schema may be connected via standard APIs to applicable agency software (be that permitting, inspection, procurement, appeals, and other applicable agency systems). In the long term, handoffs between these agencies can become connected and timestamped events that automatically record operational mechanics as day-to-day operations proceed across a state.

The long-term goal is a government that has legibility into the differences between active agency work and queue time, locates bottlenecks and dependency failures, and so on – all the while lowering the documentation requirements that burden bureaucratic processes. Rather than exchanging information through sequential document transfers, participating agencies share operational and permitting telemetry through APIs with the workflow database. This gives lawmakers real-time feedback into the operations their work creates.



Figure 10 – ***Agency API Connections:*** *Setting up inter-agency API data connections automates permitting telemetry and allows us to automate what would otherwise be inferred from siloed systems.*

7. Version-Controlled Governance

This digital structure also enables lawmakers to search for broken dependencies, critical gaps in regulatory frameworks that have been solved elsewhere and enables lawmakers to 'roll back' changes if previously unidentified problems arise. Using technology like Git for this purpose (which is already widely utilized in tech), each 'version' of the regulatory framework is automatically preserved, with its associated outcomes, so that lawmakers can track metrics of note over time.

Preserving this structure also enables greater transparency for the public, as they can observe line-by-line authorship of laws by state or federal lawmakers, which also enables a greater degree of cross-examination if lawmakers write clauses and loopholes into thousand-page documents that suspiciously exempt their donors from taxes, regulatory changes, or similar.

8. Discussion

Governments have traditionally been great at creating or repealing laws, but they have persistently lacked the capability for parsing increasingly complex administrative systems those laws create.

This proposal is a technological maintenance tool designed for this very purpose. It connects the legal texts drafted by lawmakers to the operational workflows those texts produce. It makes those workflows directly observable via dashboards, and provides governments with a disciplined pathway for piloting, evaluating, and improving regulatory frameworks over time.

If successful, this proposal would fundamentally improve the regulatory system from its current state. Legislatures and agencies would retain their authority to create laws and regulations but gain better tools and greater visibility into the operational workflows those regulations create.

With this proposal, lawmakers gain the ability to directly observe how regulations affect the construction timelines of solar farms, hospitals, houses, and other projects. They can compare their jurisdiction's regulatory environment with peer states to drive efficiency. They can easily identify agency bottlenecks and regulatory gaps. Finally, they can test changes to a regulatory framework before adopting them broadly.

The larger objective is to make regulatory systems legible as integrated software operations — currently known as DevOps. This enables continuous improvement and continuous delivery for government systems, along with a middle path between accumulated opaque regulations and indiscriminate repeal. Over time, this proposal shifts the equilibrium from a steady accumulation of regulatory layers towards a gradual adoption of the most efficient regulatory framework possible.

References

1. Fukuyama, Francis. Political Order and Political Decay: From the Industrial Revolution to the Globalization of Democracy. New York: Farrar, Straus and Giroux, 2014. pg. 514–517, 540.

2. Teles, Steven M. Kludgeocracy: The American Way of Policy. Washington, D.C.: New America Foundation, 2012. pg. 5–6.
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